

Wideband-Efficient SOI Uniform Subwavelength Grating Couplers by Effective-Index and Leakage-Factor Matching at Multiple Wavelengths

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Abstract—By improving both effective-index (EI) matching and leakage-factor (LF) matching at multiple wavelengths, a method to achieve a wideband-efficient uniform subwavelength grating coupler (SWGC) has been theoretically demonstrated. First, the root-mean-square-error between the grating EI and ideal EI (RMSE-EI) and root-mean-square-error between the grating LF and ideal LF (RMSE-LF) are adopted to weigh EI and LF matching at multiple wavelengths. It is found that when the RMSE-EI decreases and keeping RMSE-LF almost constant, the coupling bandwidth of the uniform SWGC could be substantially increased. Moreover, when RMSE-LF of the uniform SWGC decreases while RMSE-EI remains almost constant, the overall integral coupling efficiency (ICE) within the band from 1525 nm to 1575 nm is enhanced. Furthermore, by simultaneously enhancing both EI matching and LF matching at three wavelengths, a 1-dB bandwidth of 53.7 nm and ICE of 22.31 is obtained in an optimized SOI uniform SWGC. Using the proposed multi-wavelength optimizing methodology, even more efficient wideband couplers could be expected by adopting more optimization accounting for more structural parameters in the future.

Index Terms—Silicon photonics, grating coupler, multi-wavelengths matching, effective-index matching, leakage-factor matching.

I. INTRODUCTION

SILICON photonics [1], [2], [3] has undergone dramatic progress, moving concepts from academic study to industry; however, Si optical chips confront the challenge of efficient coupling between chip and fiber due to size and mode mismatch [4], [5]. The problem is generally tackled by edge coupling or grating (vertical) coupling [2]. Edge couplers can achieve coupling losses below 0.5 dB and bandwidths greater than 100 nm [6]. However, the device needs optical-quality facets on the chip sides and small alignment tolerances [7], adding complex fabrication steps and thus creating difficulties for mass production and testing [1].

Grating couplers (GCs) are more attractive as they are convenient for on-chip automatic testing and have flexible positioning tolerances [1]. To be specific, GCs enable the coupling of light propagating in a slab or ridge waveguide to optical fiber. Because numerous optical systems and subsystems are based on SOI waveguides, the GCs considered are also based on these materials, potentially simplifying integration of these GCs with other SOI devices. Light along uniform subwavelength grating couplers (SWGC) decays exponentially, and consequently it can achieve a limited theoretical coupling efficiency (CE) of 80% [4] when coupling with a Gaussian beam. By apodization [5], [6], [7], [8], adding bottom reflectors [9], and adding overlayer on the top layer [10], and adopting a blazed grating structure [11], the CE at the central wavelength can often be improved further; however, in practice the bandwidth of GCs with high efficiency rarely exceeds 50 nm [6]. Meanwhile, increased dispersion [7] and use of low refractive-index gratings [12], [13] also have been proposed to extend the bandwidth, and values in excess of 100 nm [12] have been reported, but these investigations rarely take CE into consideration simultaneously with optimizing bandwidth. In fact, there is a precise trade-off between CE and bandwidth [9]. Recently, by additionally optimizing the structure of an upper layer, a 90-nm 1-dB bandwidth was numerically demonstrated [3], in which the wavelength sensitivity to the field-pattern is found to be especially important to achieve efficient broadband GCs. It should be noted, however, that the investigation mainly focuses on structural optimization;

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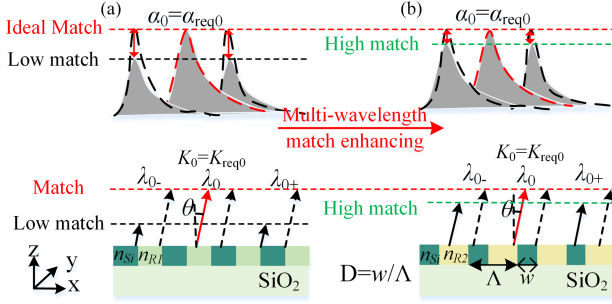


Fig. 1. Schematic of multi-wavelength matching. (a) Conventional EI and LF matching optimized to a single wavelength. (b) Multi-wavelength EI and LF matching. Dashed lines represent the required ideal grating phase vector and ideal field-pattern. In the upper panel, the actual field-pattern is shaded with grey. The parameters for the uniform SWGC are given in the main text.

how multi-wavelength field-pattern matching affects the coupling has not been explored. Moreover, the multi-wavelength effective-index (EI) matching aspects are also not addressed. If both EI and field-pattern matching are improved, wideband high CE should be expected.

Therefore, in this letter, in contrast to previous investigations that only consider matching at *single* central wavelength [5], EI and field-pattern matching at multiple wavelengths is investigated to obtain wideband and high CE. For uniform SWGC, the field-pattern matching between the grating diffracted field and the fiber Gaussian field is directly related to the leakage-factor (LF) of the gratings [14], so in the below context, LF matching is used to estimate field-pattern matching between the grating and the fiber. That is, the better the LF matching, the better the field-pattern matching. To accurately estimate the coupling performance, the integral coupling efficiency (ICE) [15] within the 1525-1575 nm band of key importance to optical communications for uniform SWGC are compared. In this work, we use the two-dimensional finite-difference time domain (2D-FDTD) method to simulate in *MEEP*, which is the same as the simulation method used in Ref. [16]. A wideband and efficient uniform SWGC with highest ICE of 22.31 is achieved, and the peak CE and 1-dB bandwidth are 49% at 1550 nm and 53.7 nm, respectively. In practice, ICE and bandwidth are the key figures of merit of interest.

II. SUBWAVELENGTH GRATING COUPLER DESIGN AND OPTIMIZATION

To achieve wideband high ICE, Fig. 1(b) shows a schematic diagram of our proposed multi-wavelength EI and LF matching, and as a comparison, conventional single-wavelength matching scheme is also shown in Fig. 1(a). According to our approach, Fig. 1(b), EI and LF matching is not only considered at the center wavelength λ_0 , high matching at two other wavelengths λ_{0+} and λ_{0-} are also required. Here, the wavelength ranges from λ_{0-} to λ_{0+} is the band in which we are interested. For EI matching, at λ_{0+} and λ_{0-} , the diffraction vector K_{0+} or K_{0-} should also be close to the required ideal vectors $K_{\text{req}0+}$ or $K_{\text{req}0-}$. Consequently, the matching error between the actual grating EI (n_{eff}) and the required ideal EI (n_{req}) at the corresponding wavelength

can be decreased by our approach. Similarly, improving LF matching at multiple wavelengths could be achieved by reducing the error between the grating leaky factor α and the required α_{req} . When α_{0+} (α_{0-}) and the required $\alpha_{\text{req}0+}$ ($\alpha_{\text{req}0-}$) are close, LF matching at λ_{0+} and λ_{0-} also can be enhanced. Therefore, the multi-wavelength matching method could improve coupling at all wavelengths considered, and thus it should increase both the coupling bandwidth and ICE.

To weigh the overall grating match, the root mean square error (RMSE) of the EI matching and the LF matching at multiple wavelengths (RMSE-EI and RMSE-LF) are introduced in (1) and (2),

$$RMSE - EI = \sqrt{\sum_{i=0}^N (n_{\text{eff}}(\lambda_i) - n_{\text{req}}(\lambda_i))^2 / N}, \quad (1)$$

$$RMSE - LF = \sqrt{\sum_{i=0}^N (\alpha(\lambda_i) - \alpha_{\text{req}}(\lambda_i))^2 / N} \quad (2)$$

where $n_{\text{eff}}(\lambda_i)$ and $n_{\text{req}}(\lambda_i)$ are the grating EI and the required ideal EI at the corresponding wavelengths [please see Supplementary information for detailed calculations of $n_{\text{eff}}(\lambda_i)$ and $n_{\text{req}}(\lambda_i)$]. In addition, $\alpha(\lambda_i)$ and $\alpha_{\text{req}}(\lambda_i)$ are the grating field leakage factor and required ideal leakage factor at the corresponding wavelengths [please see Supplementary information for detailed calculations of $\alpha(\lambda_i)$ and $\alpha_{\text{req}}(\lambda_i)$]. λ_i represents the corresponding wavelength, N is the number of wavelengths; in this paper, $N = 3$ (1525, 1550, and 1575 nm) to cover a 50-nm band [6], which should be sufficient to demonstrate our principle for more wavelengths.

Now that the principle has been illustrated, we proceed to show that it can enable the design of a wideband high-CE GC. For comparison, a state-of-the-art uniform SWGC [16] for central wavelength 1550 nm and TE polarization is adopted, as shown in the bottom of Fig. 1(a). The period of the uniform SWGC is Λ , which consists of Si with refractive index 3.45 and an artificial material with index n_{R2} [7]. The duty cycle D is defined as the length of Si to Λ . The thickness of the top Si layer is 250 nm, and that for the bottom box layer is 3 μm . To avoid second order reflection, the coupling angle is 10° , and the coupling position is $L_c = 3.8 \mu\text{m}$ (L_c is defined as the distance between the fiber core center and the beginning of the grating). During the investigation, $n_{\text{eff}}(\lambda)$ and $n_{\text{req}}(\lambda)$ at each wavelength are obtained by effective-medium theory (EMT) and phase-matched conditions [13], and $\alpha(\lambda)$ and $\alpha_{\text{req}}(\lambda)$ for the uniform SWGC are calculated by extracting the power along the grating [8].

III. RESULTS AND DISCUSSION

First, the multi-wavelength EI and LF matching degree of the coupling influenced by D and n_{R2} are explored. As shown in Fig. 2, we carried out calculations sweeping the parameters to obtain the RMSE-EI and RMSE-LF distributions for different grating structures. The RMSE-EI and RMSE-LF distributions with D between 0.1 and 0.7 and n_{R2} between 1.45 and 3.45 have been explored at a fixed $\Lambda = 806 \text{ nm}$, which is similar with a previous report [17]. From Fig. 2(a), when n_{R2} exceeds 2.5, the RMSE-EI is always high, which means a worse match. For n_{R2}

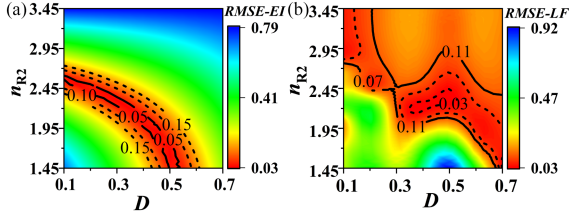


Fig. 2. Contour maps of (a) RMSE-EI and (b) RMSE-LF with various n_{R2} and D . The red and blue regions represent low and the high RMSE of EI and LF, respectively.

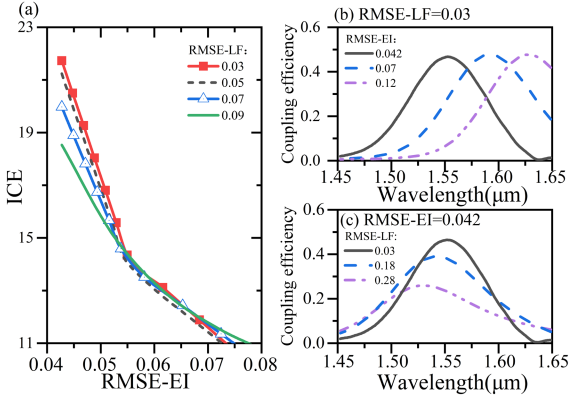


Fig. 3. (a) ICE as a function of RMSE-EI corresponding to RMSE-LF of 0.03, 0.05, 0.07 and 0.09. (b) The CE spectrum for uniform SWGC with various RMSE-EI when RMSE-LF = 0.03. (c) The CE spectrum for uniform SWGC with various RMSE-LF when RMSE-EI = 0.042.

< 2.5 , as D rises from 0.1 to 0.7, the RMSE-EI first drastically decreases to a minimum, and it then gradually increases. Meanwhile, Fig. 2(b) shows the RMSE-LF influenced by the same structural parameters corresponding to Fig. 2(a). Despite the fact that RMSE-LF is small (which means a good field-pattern match), when $n_{R2} > 2.5$, its correspondingly RMSE-EI shown in Fig. 2(a) is always high. Thus, the coupling situation with these parameters is not favorable, and thus $n_{R2} > 2.5$ should not be considered. For n_{R2} in $[1.8, 2.5]$, as D rises, RMSE-LF first decreases to a minimum, and it then gradually increases. By considering both the ranges of Fig. 2(a) and (b), the minimum RMSE-EI and minimum RMSE-LF appear in a common zone with $D \sim 0.3 \sim 0.4$ and n_{R2} within $[1.8, 2.5]$, which should be taken to improve EI and LF matching. By adjusting structure parameters, a better RMSE-EI and RMSE-LF could be obtained, but whether better RMSE-EI and RMSE-LF can really improve the overall coupling performance remains to be answered.

Therefore, how RMSE-EI and RMSE-LF affect the coupling performance needs to be investigated. To focus first on the effect of RMSE-EI, we keep RMSE-LF almost the same and only change RMSE-EI, which are obtained by taking the corresponding structural parameters from the RMSE-LF contour lines of 0.03, 0.05, 0.07 and 0.09 in Fig. 2(b) with $n_{R2} < 2.5$. As shown in Fig. 3(a), at each RMSE-LF level, ICE decreases while RMSE-EI increases, which supports our previous expectation that better RMSE-EI really improves the coupling performance. To specifically analyze how RMSE-EI affects CE, Fig. 3(b) shows the coupling spectra within the target wavelength range for various RMSE-EI with RMSE-LF fixed at 0.03. In Fig. 3(b),

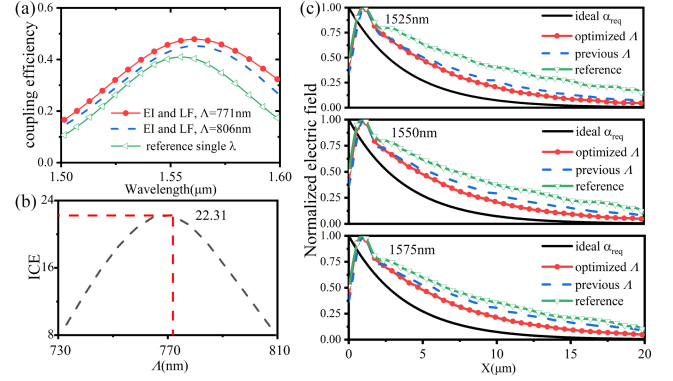


Fig. 4. (a) Comparison of CE spectra for different optimized uniform SWGC. (b) The ICE as a function of λ for $D = 0.4$ and $n_{R2} = 2.2$. (c) The normalized electric field E_x profiles comparisons for coupling-out uniform SWGC at 1525 nm, 1550 nm and 1575 nm, corresponding to the different optimized uniform SWGC in Fig. 4(a), as well as the ideal leakage factor α_{req} distribution curve.

as RMSE-EI increases from 0.042 to 0.12, while the peak wavelength gradually increases and deviates from the wavelength range of interest, the peak CE of the spectrum remains almost constant. This indicates that RMSE-EI has a heavy influence on the matching spectrum, but little influence on peak CE.

Next, we turn our attention to the effect of RMSE-LF. As shown in Fig. 3(a), when RMSE-EI is below 0.055, with a smaller RMSE-LF, a larger ICE could be obtained as expected, which is normal phenomenon. However, when RMSE-EI is above 0.055, with a smaller RMSE-LF, the obtained ICE is also smaller, which is contrary to our expectations. This is due to the fact that a constant ideal $\alpha_{req} = 0.13$ is adopted in our calculation, which is an optimum only valid for 1550 nm at which wavelength perfectly EI match is satisfied [18]. Therefore, our calculation of RMSE-LF related to improvement of field-pattern matching is only valid when EI matching is approaching perfect, for the case that RMSE-EI is below 0.055. Otherwise, when RMSE-EI gradually increases, resulting in poor EI matching that makes the optimal coupling wavelength deviate from our band of interest, when the ideal α_{req} deviates from 0.13, an improvement of RMSE-LF will lead to little improvement of field-pattern matching. Thus, the unexpected phenomenon is displayed. In Fig. 3(c), holding RMSE-EI constant, for example 0.042, the effect of varying RMSE-LF on the CE is investigated. As RMSE-LF increases from 0.03 to 0.28, the wavelength that maximizes CE shifts to shorter wavelength, and the peak value of CE rapidly decreases. It indicates that RMSE-LF has a strong influence on both the coupling spectrum and the peak CE. Thus, RMSE-LF should be treated more carefully to obtain high CE with a broad bandwidth.

Having confirmed the effectiveness of RMSE-EI and RMSE-LF in enhancing uniform SWGC performance, utilizing the combined effect of RMSE-EI and RMSE-LF is next considered. With fixed $\lambda = 806$ nm, by optimizing D and n_{R2} , RMSE-EI and RMSE-LF are simultaneously improved. Thus, the maximum ICE of 21.44 for the uniform SWGC is achieved with $D = 0.4$ and $n_{R2} = 2.2$. Its corresponding CE spectrum is shown in Fig. 4(a), labeled by the blue dashed curve. To obtain better coupling, we consider varying λ to further improve RMSE-EI

TABLE I
COMPARISON OF COUPLING PERFORMANCE BETWEEN DIFFERENT SWGCs

Ref	SWGC types	BW, nm	Peak CE (dB)	Upper cladding or bottom reflector	Box, μm
[10]	With upper cladding	75 ¹ dB	-1	both with	1.26
[12]	Focusing	100 ¹ dB	-4.7	with upper cladding	3
[11]	Blazed	35 ¹ dB	-0.46	with bottom reflector	2.43
[19]	Uniform	40 ¹ dB	-4.32	without	1
[16]	Uniform	37 ¹ dB	-3.76	without	1
[20]	Apodized	55 ³ dB	-3.1	without	3
This work	Uniform	53.7 ¹ dB	-3.09	without	3

and RMSE-LF. This time, the previously optimized $n_{R2} = 2.2$ and $D = 0.4$ are fixed, and Λ is allowed to vary from 730 to 810 nm, in this range, the EI matching and LF matching show good performance, as shown in Fig. 4(b). A maximum ICE of 22.31 is obtained with optimized $\Lambda = 771$ nm, and its corresponding CE spectrum is also shown in Fig. 4(a), labeled with red solid curve with the circles. The optimized uniform SWGC has a peak CE of 3.09 dB (49%) and 1-dB bandwidth of 53.7 nm (83.43 nm at 3 dB). In addition, as a comparison, the CE spectrum of a previous reported uniform SWGC [16], which is optimized only for the single center wavelength of 1550 nm, is also plotted in Fig. 4(a) as the green curve with hollow triangles. With the structural parameters supplied by the reported uniform SWGC [16], the RMSE-EI and RMSE-LF we calculated are 0.17 and 0.07, respectively. They are larger than our multi-wavelength optimized RMSE-EI and RMSE-LF, and its ICE is only 18.4, which is lower than that for our optimized uniform SWGC. From Fig. 4(a), it is obvious that reducing RMSE-EI and RMSE-LF can effectively improve ICE.

To better understand the effect of multi-wavelength coupling, Fig. 4(c) shows the normalized electric field E_x distribution corresponding to the various optimized uniform SWGC in Fig. 4(a) at the three wavelengths 1525, 1550, and 1575 nm. The field distributions for an ideal leakage factor α_{req} of 0.13 at these wavelengths are also plotted as a reference [18]. As shown in Fig. 4(c), while the normalized E_x distribution of the optimized multi-wavelength uniform SWGC with $\Lambda = 771$ nm is closest to the field distribution of the ideal α_{req} , the normalized E_x distribution of the previous reported uniform SWGC [16] deviates the most from the field distribution of the ideal α_{req} . It indicates that with the decrease of RMSE-LF, the normalized E_x distribution of the grating diffraction field at corresponding wavelength gradually approaches the field distribution of the ideal α_{req} , showing good field-pattern matching, which results in an increase of CE. On the contrary, with the increases of RMSE-LF, the normalized E_x distribution of the grating diffraction field at corresponding wavelength gradually deviates from the field distribution of the ideal α_{req} , showing poor field-pattern matching, which results in a decrease of CE.

As shown in Table I, we compare the coupling performance of other reported SWGCs with that of our designed SWGC.

Compared with other SWGCs in Refs. [16], [19] and [20], our SWGC shows better coupling performance for similar structural complexity. In the structure without the additional upper cladding layer, the BW of our SWGC is lower than that of Refs. [10] and [12], and for that without the additional bottom reflector, the CE of our SWGC is also lower than that of Refs. [10] and [11]. We also investigated the fabrication tolerances of Λ and D . In the case of grating parameters $D = 0.4$, $\Lambda = 771$ nm, and $n_{R2} = 2.2$, to ensure the good coupling performance of SWGC, it is best to control the fabrication tolerances of Λ and D within 10nm and 9.25nm, respectively.

IV. CONCLUSION

In conclusion, to obtain wideband and efficient uniform SWGC, a multi-wavelength coupling matching method between optical fiber and a GC by considering RMSE-EI and RMSE-LF has been proposed. After investigation, we find that, first, decreasing both RMSE-EI and RMSE-LF improves ICE. Second, by improving RMSE-EI and RMSE-LF, the ICE can be enhanced to 21.44, which is a significant improvement compared to previous single-wavelength uniform SWGC [16] with an ICE of 18.4. Moreover, by further improvement of RMSE-EI and RMSE-LF through additionally adjusting Λ , ICE as high as 22.31 is achieved. Both of the above cases clearly demonstrate that using the proposed multi-wavelength matching method, ICE could be further improved by introducing more structural degrees of freedom. Furthermore, although both RMSE-EI and RMSE-LF can affect the ICE, their effects are different. RMSE-EI has a strong influence on the matching spectrum, but little influence on peak CE; RMSE-LF has a strong influence on the peak CE and a moderate influence on the matching spectrum. These findings provide a useful methodology for the design of other types of wideband and efficient GCs. Because of the exponentially decaying diffraction field of the uniform SWGC, the best match with the Gaussian field of the fiber is limited. A promising solution is to use the multi-wavelength matching method in designing apodized GCs, which can adjust the diffraction field of the GC to approach a Gaussian shape, thus improving the CE of GC at multiple wavelengths. Finally, we remark that it is promising to apply this methodology to silicon nitride and lithium niobate platforms, which are more suitable for designing broadband SWGC, and with greater fabrication tolerances.

See the supplementary material for the supporting contents: 1. Detailed explanation of each abbreviation in the text. 2. Detailed description of design principles and calculation methods.

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